CP 218 Theory and Applications of Bayesian Learning

Project 2: Bayesian Data Analysis

Yakshith Maddineni (ID: 26080)

1 Introduction

This project presents a comprehensive **Bayesian analysis** of an Indian road accident dataset spanning the years **2022–2025**, employing advanced probabilistic modeling techniques to generate deeper insights into road safety patterns. By evaluating **Bayesian Regression**, **Multinomial Logistic Classification**, and **Unsupervised Mixture Models**, the study captures complex relationships among **geospatial**, **temporal**, and **environmental factors**. The integration of **high-performance computing** through **JAX-based optimization** and **NUTS sampling** enables robust uncertainty quantification and improved predictive reliability when compared to traditional frequentist baseline approaches. Overall, the findings demonstrate the effectiveness of **hierarchical** and **generative Bayesian frameworks** in identifying high-risk accident patterns and uncovering latent traffic structures.

2 Dataset Overview

The dataset spans eight major Indian cities from 2022 to 2025 and integrates diverse feature sets for predictive modeling. It includes *accident and geospatial features*, such as severity levels, number of vehicles involved, and casualty counts linked to specific coordinates, cities, and states. In addition, *temporal features* capture time-series variables including hour, day, weekend indicators, and peak-hour flags. The dataset also incorporates *environmental and road-related attributes*, such as weather conditions, visibility, road type, lane count, and traffic density. Furthermore, *contextual features* include festival indicators and an engineered *Risk Score* ranging from 0 to 1, designed to support advanced road safety research and machine learning applications. The dataset is publicly available on Kaggle at `https://www.kaggle.com/datasets/sehaj1104/indian-road-accident-dataset-20222025`.

3 Bayesian Regression

A Bayesian regression framework was developed to model the road accident risk score using three probabilistic models: Gaussian Linear Regression, Beta Regression, and Hierarchical Beta Regression. To evaluate their effectiveness, the models were compared in terms of predictive accuracy, convergence diagnostics, and posterior reliability. The analysis focused on the Bayesian framework's ability to capture uncertainty, ensure predictions remain bounded between 0 and 1, and improve robustness through hierarchical shrinkage priors, which is particularly important for reliable accident risk assessment under complex and heterogeneous road conditions.

3.1 Model Summary

The Bayesian regression analysis was carried out using three models. **Model 1**, the Gaussian Linear Regression model, was used as the baseline approach and assumes a linear relationship between the predictor variables and the accident risk score through a Gaussian likelihood. **Model 2**, the Beta Regression model, extends this approach by introducing a non-linear sigmoid link function, ensuring that the predicted risk scores remain bounded between 0 and 1, which is more suitable for fractional response data. **Model 3**, the Hierarchical Beta Regression model, further improves the framework by incorporating feature-dependent precision along with hierarchical shrinkage priors, allowing better handling of variability and improving robustness in the presence of complex relationships among predictors.

3.2 Sampling and Execution Configuration

The models were executed using the **NUTS** sampler through **NumPyro** with **JAX**-based XLA optimization for accelerated computation. Each model was run with **2–4 independent chains**, using **1,000–2,000 tuning steps** followed by **1,000–2,000 posterior draws**. To ensure sampling stability in the Beta regression models, the `target_accept` parameter was maintained between **0.90 and 0.98**. Reproducibility was ensured using a fixed random seed (`42`), while convergence was assessed through **R-hat** values close to **1.0** and adequate **Effective Sample Size (ESS)**.

3.3 Prior Distributions

The prior distributions used across the three Bayesian models. The first model uses weakly informative priors to allow greater flexibility in parameter estimation. The second model adopts tighter informative priors to improve

regularization and reduce overfitting. The third model introduces hierarchical shrinkage priors, which allow the model to learn the appropriate amount of regularization directly from the data. This hierarchical structure improves model stability and handles feature uncertainty more effectively.

Model	**Prior Type**	**Specification**
Model 1	Weakly Informative	$\beta \sim N(0, 10), \alpha \sim N(0, 10)$
Model 2	Informative	$\beta \sim N(0, 1), \alpha \sim N(0, 2)$
Model 3	Hierarchical Shrinkage	$\tau \sim \text{HalfNormal}(1), \beta \sim N(0, \tau)$

Table 1: Prior Choices Used in Each Model

3.4 Convergence Diagnostics

Model 1 shows poor convergence with a high $\hat{R}$ value and a very low effective sample size, indicating instability in posterior sampling. Model 2 shows improved convergence with stable posterior estimates and reliable inference. Model 3 demonstrates excellent convergence with $\hat{R} = 1.00$ and a very high effective sample size, confirming that the posterior samples are reliable and well-mixed.

Model	$\hat{R}$	**Bulk ESS**	**Interpretation**
Model 1	up to 2.90	2 – 621 (many < 10)	Poor mixing and unstable posterior estimates
Model 2	1.00	∼4,000 – 10,000+	Good convergence and reliable inference
Model 3	1.00	$> 9,000$ (many $> 20,000$)	Excellent convergence with well-mixed chains

Table 2: Convergence diagnostics comparison of the three Bayesian models

3.5 Predictive Performance

The predictive performance of the three models using R^2 and Mean Absolute Error (MAE). Although the Gaussian regression model achieves the highest R^2, its poor convergence makes the results less reliable. The hierarchical beta regression model provides a strong balance between predictive accuracy and model reliability. Therefore, it is considered the most suitable model for practical deployment and interpretation.

Model	$\mathbf{R^2}$	**MAE**
Model 1	0.891	0.050
Model 2	0.872	0.060
Model 3	0.881	0.055

Table 3: Predictive Performance on Test Data

3.6 Interpretation of Results

Although Model 1 achieved the highest nominal R^2 score, it failed convergence diagnostics and therefore its posterior estimates are unreliable. Model 3 achieved strong predictive performance with excellent convergence statistics, making it the most reliable and practically useful model.

3.7 Model Comparison

Among the three Bayesian models, the **Hierarchical Beta Regression** model is the most suitable because it ensures that predictions remain bounded between 0 and 1, which is highly appropriate for accident risk scores. It employs hierarchical shrinkage priors that automatically regularize the regression coefficients, thereby reducing overfitting and improving model stability. Furthermore, the model captures feature-dependent uncertainty, enabling it to better represent variations and heteroskedasticity present in the data. In addition, it demonstrates excellent convergence diagnostics, with $\hat{R} = 1.00$, a high effective sample size (ESS), and zero divergences. These properties make its predictions both accurate and statistically reliable when compared with the other two models.

4 Bayesian Classification

A Bayesian Multinomial Logistic Regression model was developed to classify road accident severity across three levels: *Minor* (0), *Major* (1), and *Fatal* (2). To evaluate its efficacy, the model was benchmarked against a Gaussian Naive Bayes baseline. The analysis focused on the Bayesian framework's ability to improve predictive accuracy, manage class imbalance, and utilize posterior probability estimates for uncertainty quantification, which is critical for identifying high-risk fatal cases.

4.1 Model Summary

Two contrasting models were evaluated on the accident severity dataset. The baseline, *Gaussian Naive Bayes (GNB)*, is a generative frequentist classifier that assumes conditional independence among predictors, expressed as $P(y \mid X) \propto P(y) \prod_{j=1}^{p} P(x_j \mid y)$, where the predictors are modeled as $x_j \mid y = k \sim \mathcal{N}(\mu_{jk}, \sigma_{jk}^2)$. This model does not incorporate explicit regularization or prior distributions. In contrast, the primary model is a discriminative Bayesian Multinomial Logistic Regression model, which directly estimates class probabilities using the softmax function, $P(y = k \mid X) = \frac{\exp(\eta_k)}{\sum_{c=1}^{K} \exp(\eta_c)}$, where $\eta_k = \beta_k^T X + \alpha_k$. Unlike the frequentist baseline, posterior inference was performed using Markov Chain Monte Carlo (MCMC) through the No-U-Turn Sampler (NUTS), enabling robust parameter estimation and full uncertainty quantification.

4.2 Sampling and Execution Configuration

The model was executed using the **NUTS (No-U-Turn Sampler)** with a **NumPyro** backend powered by **JAX**-based XLA optimization for accelerated computation. To ensure reproducibility, a fixed random seed (`42`) was used throughout all stochastic processes. The MCMC configuration employed **2 independent chains**, each with **1000 tuning steps** followed by **1000 posterior draws**, resulting in **2000 total samples** for inference. For stable posterior exploration, the `target_accept` parameter was set to **0.95**, reducing divergent transitions and improving sampling accuracy.

4.3 Prior Distributions

The Bayesian model employs Normal priors for both the coefficient and intercept parameters, defined as $\beta \sim \mathcal{N}(0, \sigma)$ and $\alpha \sim \mathcal{N}(0, \sigma)$. To assess the robustness of posterior inference and predictive performance, three prior scales were investigated through sensitivity analysis.

Prior Type	σ	Interpretation
Weakly Informative	1	Mild regularization
Medium	2	Moderate flexibility
Non-informative	5	Data-dominant inference

Table 4: Prior Sensitivity Configurations

4.4 Convergence Diagnostics

Convergence of the Bayesian Multinomial Logistic Regression model was evaluated using the Gelman–Rubin statistic ($\hat{R}$) and effective sample size (ESS). Lower $\hat{R}$ values close to 1 and higher ESS values indicate better chain convergence and stable posterior sampling.

Prior	$\hat{R}$	Bulk ESS	Tail ESS	Interpretation
$\sigma = 1$	≤ 1.0025	> 3900	> 1060	Excellent
$\sigma = 2$	≤ 1.0012	> 1100	> 1260	Good
$\sigma = 5$	1.0038–1.0118	109–184	255–299	Poor Mixing

Table 5: MCMC Convergence Diagnostics

The weakly informative prior ($\sigma = 1$) demonstrated the most stable convergence behavior, indicating superior posterior mixing and reliable parameter estimation compared to broader priors.

4.5 Predictive Performance

The predictive performance of the baseline and Bayesian models was evaluated using multiple classification metrics, including accuracy, weighted precision, weighted recall, weighted F1-score, macro F1-score, and Fatal-class F1-score. The results indicate that all Bayesian prior settings consistently outperformed the Gaussian Naive Bayes baseline, particularly in macro-level performance and fatal accident classification.

Model	Accuracy	W-Precision	W-Recall	W-F1	Macro F1	Fatal F1
Naive Bayes	0.5298	0.4684	0.5298	0.4737	0.4233	0.50
Bayesian ($\sigma = 1$)	0.5585	0.5978	0.5585	0.5683	0.6472	1.00
Bayesian ($\sigma = 2$)	0.5613	0.6008	0.5613	0.5710	0.6494	1.00
Bayesian ($\sigma = 5$)	0.5635	0.6022	0.5635	0.5732	0.6509	1.00

Table 6: Model Performance Comparison

The Bayesian models achieved substantial improvements over the baseline, with near-identical performance across different prior scales, suggesting that predictive accuracy is relatively insensitive to prior variance selection.

4.6 Prior Sensitivity Results

The effect of prior scale on predictive performance was found to be minimal, with only marginal differences observed across accuracy and F1-score metrics. This indicates that the Bayesian model is robust to prior specification and produces stable predictions across different prior assumptions.

Prior	Accuracy	Weighted F1	Macro F1
$\sigma = 1$	0.5585	0.5683	0.6472
$\sigma = 2$	0.5613	0.5710	0.6494
$\sigma = 5$	0.5635	0.5732	0.6509

Table 7: Sensitivity Analysis Across Priors

4.7 Model Comparison

The Gaussian Naive Bayes baseline underperforms primarily because it relies on the strong (and often unrealistic) assumption of conditional feature independence, which is violated in this correlated road-accident dataset. Its limited SMOTE strategy—oversampling only the fatal class—fails to balance the major-injury class, resulting in poor discrimination (macro F1 = 0.4233, fatal F1 = 0.50) and low overall accuracy (0.5298). In contrast, the Bayesian Multinomial Logistic Regression model succeeds by directly modelling class probabilities through the softmax function, which explicitly captures competition among the three severity levels. Combined with SMOTE-Tomek balancing of all classes and weakly informative priors that provide appropriate regularization, it achieves substantially higher predictive performance (macro F1 = 0.6472, fatal F1 = 1.00, accuracy = 0.5585) while delivering well-calibrated uncertainty quantification. This demonstrates that the discriminative Bayesian approach is far more suitable for this multi-class, imbalanced problem than the generative Naive Bayes classifier.

5 Bayesian Unsupervised Learning

A Bayesian mixture modeling framework was developed to segment road traffic data into three latent clusters using both a Dirichlet Process Gaussian Mixture baseline and a fully Generative Bayesian Mixture Model. The analysis focused on identifying hidden traffic patterns, improving cluster separation, and quantifying uncertainty through posterior inference, enabling reliable detection of high-risk temporal and road-condition segments.

5.1 Model Summary

Two Bayesian Mixture Model variants were implemented to segment the road traffic data into $K = 3$ clusters. The first, **Bayesian Gaussian Mixture Model (BGMM) (Baseline)**, uses a truncated Dirichlet Process prior with `weight_concentration_prior` = 0.01 to automatically suppress unnecessary clusters while assuming full covariance matrices for flexible cluster modeling. The second, **Generative Bayesian Mixture** , is a fully probabilistic PyMC framework with a Dirichlet prior on cluster weights and Normal/HalfNormal priors on component parameters. Posterior inference was performed using **NUTS** via **NumPyro**.

5.2 Sampling and Execution Configuration

Models were executed on a **JAX-optimized backend** with **GPU acceleration (CUDA/XLA)** to efficiently handle the **20,000-sample dataset**. Posterior inference was performed using the **No-U-Turn Sampler (NUTS)** with **two vectorized chains**, each comprising **1,000 tuning steps** and **1,000 posterior draws**. To ensure stable convergence and minimize divergent transitions, the **target acceptance rate** was fixed at **0.90**.

5.3 Convergence Diagnostics

Convergence of the Generative Bayesian mixture model was assessed using the Gelman Rubin diagnostic statistic ($\hat{R}$) along with trace plot analysis. Although the sampling process was computationally efficient, achieving approximately **8.06 iterations per second**, the $\hat{R}$ values for certain parameters exceeded the recommended threshold of 1.01. In addition, the effective sample size (ESS) for some parameters was observed to be below 100. These indicators suggest the presence of **label switching**, a well-known phenomenon in Bayesian mixture modeling in which cluster labels interchange across different chains during posterior sampling. This behavior reflects the symmetry of mixture components and does not necessarily indicate model failure or poor posterior exploration.

5.4 Predictive Performance

The clustering quality of both implementations was assessed using the **Silhouette Score**, which evaluates the cohesion within clusters and the separation between clusters. The **Generative Bayesian Mixture Model** achieved a slightly higher Silhouette Score than the **BGMM**, indicating better cluster separation and stronger intra-cluster cohesion. Furthermore, the cluster weight distribution obtained from the Generative Bayesian Mixture model is comparatively more balanced, suggesting a more stable and representative latent structure in the road traffic data.

Model Implementation	Silhouette Score	Cluster Weight Distribution
BGMM	0.37	[0.210, 0.206, 0.584]
Generative Bayesian Mixture	0.40	[0.313, 0.186, 0.500]

Table 8: Comparison of Clustering Performance

5.5 Interpretation of Clusters

The **Generative Bayesian Mixture Model** outperformed the baseline implementation by achieving a higher Silhouette Score of **0.4**, compared to **0.37** for the Scikit-Learn BGMM, indicating more distinct and well-separated clusters. While both models consistently identified **low visibility** as a pervasive underlying factor across the dataset, it produced a more practically meaningful temporal segmentation. In particular, it effectively isolated the **7:00 PM evening rush period** on **high-capacity roads**, highlighting a critical peak-risk traffic window.

Cluster	Hour	Lanes	Visibility
0	12.0	3.0	low
1	19.0	4.0	low
2	4.0	3.0	low

Table 9: Cluster Summary

5.6 Model Comparison

The **Generative Bayesian Mixture Model** is identified as the preferred approach for this analysis. While the **BGMM baseline** provided a faster initial approximation of the latent cluster structure, the **PyMC/NUTS framework** delivered a more robust estimation of cluster weights and component means, resulting in an improved Silhouette Score of **0.40**. The integration of **JAX-based XLA optimization** enabled efficient high-dimensional posterior sampling on a dataset containing **20,000 records**, which would otherwise be computationally expensive using conventional CPU-based methods. Although **label switching** was observed during sampling, the model successfully captured the underlying latent structure of the traffic data, effectively distinguishing road conditions based on **time-of-day patterns** and **infrastructure capacity**.

6 Conclusion

The study concludes that **hierarchical Bayesian models** and **generative mixture frameworks** offer superior reliability for road safety analysis by effectively handling data uncertainty and class imbalance. While traditional baseline models such as **Gaussian Naive Bayes** often struggle to capture complex feature dependencies, the Bayesian approach provides well-calibrated risk scores and more accurate severity classifications. Furthermore, by leveraging **JAX-based optimization**, the analysis demonstrates that high-dimensional traffic data can be processed with rigorous statistical convergence and efficient posterior inference. These findings establish a robust foundation for **targeted infrastructure planning**, **risk mitigation strategies**, and **policy interventions** aimed at improving road safety outcomes.

References

[1] *A Beta Regression Model: Theory and Applications*

[2] *Bayesian Multinomial Logistic Regression for Numerous Categories*

[3] *Bayesian Sparse Gaussian Mixture Model for Clustering in High Dimensions*